



TRIOS CYBER

LABORATORY PRACTICE 25

Brute-Force Style Detection Simulation

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1. Laboratory Objective and Scope

The activity simulated repeated authentication failures against the Windows laboratory account in a controlled manner. The investigation verified the corresponding Windows Security telemetry, confirmed reception in Wazuh, counted the failed attempts, and assessed whether the observed pattern would require escalation.

Element	Observed value	Purpose / result
Monitored endpoint	Windows-Agent / ID 002 / 10.10.20.30	Endpoint analyzed in Wazuh
Target account	matthew	Own laboratory account used for safe testing
Authentication event	Windows Security Event ID 4625	Failed logon telemetry
Controlled attempts	5	Repeated failures used to create the test pattern

2. Account Lockout Safety Check

Before generating failures, the local account policy was reviewed with net accounts. The captured configuration showed the account lockout threshold as Never, allowing the controlled test to proceed without intentionally triggering an account lockout.

```

matthew@trioscyber: ~
Administrador: C:\Windows\se
Microsoft Windows [Versión 10.0.19045.2006]
(c) Microsoft Corporation. Todos los derechos reservados.

matthew@DESKTOP-VOJCDSO C:\Users\matthew>net accounts
Tiempo antes del cierre forzado:          Nunca
Duración mín. de contraseña (días):       0
Duración máx. de contraseña (días):       42
Longitud mínima de contraseña:            0
Duración del historial de contraseñas:     Ninguna
Umbral de bloqueo:                        Nunca
Duración de bloqueo (minutos):             30
Ventana de obs. de bloqueo (minutos):     30
Rol del servidor:                          ESTACION DE TRABAJO
Se ha completado el comando correctamente.

matthew@DESKTOP-VOJCDSO C:\Users\matthew>

```

3. Controlled Authentication Failure Generation

Five authentication attempts were issued against the local IPC\$ resource using the matthew account and a deliberately invalid password. Each attempt returned system error 1326, confirming that the username/password combination was rejected.

```

matthew@DESKTOP-VOJCDSO C:\Users\matthew>net use \\127.0.0.1\IPC$ /user:DESK
TOP-VOJCDSO\matthew WrongPasswordDay25
Error de sistema 1326.

El nombre de usuario o la contraseña no son correctos.

matthew@DESKTOP-VOJCDSO C:\Users\matthew>net use \\127.0.0.1\IPC$ /user:DESK
TOP-VOJCDSO\matthew WrongPasswordDay25
Error de sistema 1326.

El nombre de usuario o la contraseña no son correctos.

matthew@DESKTOP-VOJCDSO C:\Users\matthew>net use \\127.0.0.1\IPC$ /user:DESK
TOP-VOJCDSO\matthew WrongPasswordDay25
Error de sistema 1326.

El nombre de usuario o la contraseña no son correctos.

matthew@DESKTOP-VOJCDSO C:\Users\matthew>net use \\127.0.0.1\IPC$ /user:DESK
TOP-VOJCDSO\matthew WrongPasswordDay25
Error de sistema 1326.

El nombre de usuario o la contraseña no son correctos.

matthew@DESKTOP-VOJCDSO C:\Users\matthew>net use \\127.0.0.1\IPC$ /user:DESK
TOP-VOJCDSO\matthew WrongPasswordDay25
Error de sistema 1326.

El nombre de usuario o la contraseña no son correctos.

```

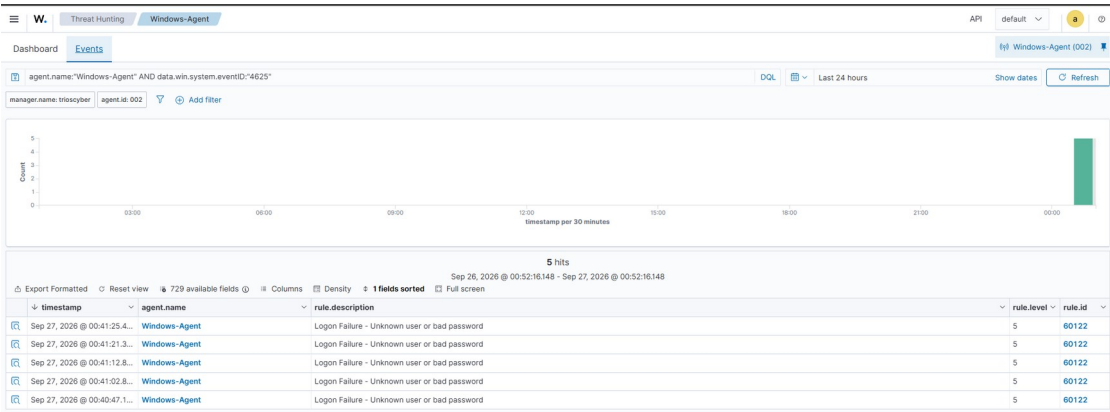
4. Windows Security Event Verification

The Windows Security log was queried for failed-logout telemetry. Event ID 4625 records a failed account login and the captured records identify matthew as the failed target account. The log evidence also shows an incorrect-credential condition and local source address 127.0.0.1.



5. Wazuh Detection and Attempt Count

Wazuh Threat Hunting displayed five matching failed-logout events from Windows-Agent. The filtered event list showed the same rule description for the five records, providing the count required for the simulation.



Metric	Observed value	Interpretation
Failed attempts	5	Five repeated authentication failures were observed
Agent	Windows-Agent / 10.10.20.30	Source endpoint identified
Event ID	4625	Windows failed-logout event
Target account	matthew	Same account targeted repeatedly

6. User-Focused Filtering

The Wazuh search was narrowed to the target account so the repeated failures could be evaluated as one short authentication pattern rather than unrelated endpoint activity.

7. Event Detail Analysis

Document Details confirmed the event context for the current failed-logout activity. The Wazuh record identifies Windows-Agent, Event ID 4625, target user matthew, source address 127.0.0.1, network logon type 3, NTLM authentication, and the status/substatus associated with invalid credentials.

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Table	JSON
index	wazuh-alerts-4.x-2026-09-27
agent.id	002
agent.ip	10.10.20.30
agent.name	windows-agent
data.vin.eventdata.authentication.packageName	NTLM
data.vin.eventdata.failureReason	NtLsa
data.vin.eventdata.ipaddress	127.0.0.1
data.vin.eventdata.ipport	49037
data.vin.eventdata.keylength	0
data.vin.eventdata.logonProcessName	NtLsa
data.vin.eventdata.logonType	3
data.vin.eventdata.processId	0x0
data.vin.eventdata.status	0xc000006d
data.vin.eventdata.substatus	0xc000006d
data.vin.eventdata.subjectLogonId	0x0
data.vin.eventdata.subjectUserSid	S-1-0-0
data.vin.eventdata.targetDomainName	DESKTOP-VJUC90
data.vin.eventdata.targetUserName	matthew
data.vin.eventdata.targetUserSid	S-1-0-0

8. Wazuh Rule and Correlation Details

Field	Observed value	Meaning
Rule ID / level	60122 / level 5	Failed-logon alert classification
Description	Logon Failure - Unknown user or bad password	Invalid credential activity identified
rule.firedtimes	5	Selected event displays a fired-times value of 5
Rule group	windows, windows_security, authentication_failed	Authentication failure classification
MITRE mapping	T1531 - Account Access Removal	Mapping preserved by the Wazuh alert

9. Escalation Assessment and First Triage Action

Escalation assessment: The five failures against the same account in a short investigation window form a brute-force-style pattern. Because the activity was intentionally generated from the local endpoint as an authorized laboratory simulation, no real incident escalation is required. An equivalent unexplained pattern in a production environment should be escalated for further investigation.

Recommended first triage action: Verify the target account, source IP, logon type, number and timing of failures, and determine whether a successful authentication occurred after the failed attempts.

10. Validation Summary

Requirement	Observed result	Status
Safe repeated failures	5 controlled invalid-password attempts against matthew	Completed
Endpoint telemetry	Windows Security Event ID 4625	Verified
Central detection	Five matching Wazuh failed-logon alerts	Verified
Attempt count	5	Confirmed
Escalation decision	Authorized lab: no real escalation; unexpected production pattern: escalate	Completed

11. Conclusions

The brute-force-style detection simulation was completed successfully. Five controlled authentication failures were generated against the matthew laboratory account and Windows recorded failed-logon telemetry under Event ID 4625. Wazuh received the events from Windows-Agent and identified the activity with rule 60122 at level 5. The filtered Wazuh view contained five matching events, and the selected event displayed rule.firedtimes = 5. The pattern is consistent with behavior that warrants investigation when unexplained; however, because the activity was authorized and locally generated for this laboratory, no real incident escalation was required.